Lecture Notes

Basic Algebraic Structures

Appendix - *Integers*

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CHAPTER

ONE

APPENDIX I - INTEGERS

1.1 Divisibility

In this chapter, our main goal is to obtain division algorithm. To achieve this we need an important consequence of the induction postulate, known as the Well Ordering Theorem.

Lemma 1.1 (Least Positive Integer) .
The integer 1 is the least positive integer.

Theorem 1.1 (The Well Ordering Principle) .
Every nonempty set S of positive integers contains a least element. That is, there is an element $m \in S$ such that $m \leq x$ for all $x \in S$.

Proof. Let S be a nonempty set of positive integers. If $1 \in S$, then $1 \leq x$ for all $x \in S$, by Lemma 1.1.

Consider now the case where $1 \notin S$, and let L be the set of all positive integers p such that $p < x$ for all $x \in S$. That is,

$$L = \{p \in \mathbb{Z}^+ \mid p < x,\ \forall x \in S\}$$

Since $1 \notin S$, Lemma 1.1 assures us that $1 \in L$. We shall show that there is a positive integer p_0 such that $p_0 \in L$ and $p_0 + 1 \notin L$. Suppose this is not the case. Then we have that $p \in L$ implies $p + 1 \in L$ (since $(\neg a \vee b) \iff (a \implies b)$). Now, since $1 \in L$, and $p \in L$ implies $p + 1 \in L$, then by Induction Postulate $L = \mathbb{Z}^+$. But this contradicts the fact that S is nonempty (note that $L \cap S =$, by definition of L). Therefore there is a p_0 such that $p_0 \in L$ and $p_0 + 1 \notin L$. Now, we must show that $p_0 + 1 \in S$. We have $p_0 < x$ for all $x \in S$, hence, $p_0 + 1 \leq x$ for all $x \in S$. If $p_0 + 1 < x$ were always true, then $p_0 + 1$ would be in L. Hence, $p_0 + 1 = x$ for some $x \in S$ and $m = p_0 + 1$ is the required least element in S. ■

Def 1.1 (Divisor) .
Let a and b be integers. We say that a **divides** b if there is an integer c such that $b = ac$. If a divides b, we write $a \mid b$.

Theorem 1.2 .
The only divisors of 1 are 1 and -1.

Proof. Suppose a is a divisor of 1. Then $1 = ac$ for some integer c. The equation $1 = ac$ requires $a \neq 0$, so either $a \in \mathbb{Z}^+$ or $-a \in \mathbb{Z}^-$. Consider first the case where $a \in \mathbb{Z}^+$. Then this requires $c \in \mathbb{Z}^+$. So we have $1 \leq a$ and $1 \leq c$, by Lemma 1.1. Then,

$$\begin{aligned} 1 < a &\implies 1 \cdot c < a \cdot c \\ &\implies c < 1 \qquad\qquad \text{(since } ac = 1\text{)}. \end{aligned}$$

But this contradicts with that $1 \leq c$. Thus, $1 = a$ is the only possibility when $a \in \mathbb{Z}^+$.

Similar arguments can be done for $-a \in Z^-$.

Therefore, combining the cases where $a \in \mathbb{Z}^+$ and where $-a \in \mathbb{Z}^-$, we have shown that either $a = 1$ or $a = -1$ if a is a divisor of 1. ■

Theorem 1.3 (The Division Algorithm) .
Let a and b be integers with $b > 0$. Then there exist unique integers q and r such that

$$a = bq + r \text{ with } 0 \leq r < b.$$

Proof. (Existence). Let S be the set of all integers x that can be written in the form $x = a - bn$ for $n \in \mathbb{Z}$, and let S' denote the set of all nonnegative integers in S. Then the set S' is nonempty. If $0 \in S'$, we have $a - bq = 0$ for some q, and hence $a = bq + 0$. If $0 \notin S'$, then by Well Ordering Principle (1.1) S' contains a least element $r = a - bq$, and hence $a = bq + r$, where $r > 0$. Now,

$$\begin{aligned} r - b &= a - bq - b \\ &= a - b(q + 1), \end{aligned}$$

and hence $r - b \in S$. Since r is the least element in S' and $r - b < r$, then $r - b$ must be negative. That is, $r - b < 0$, hence $r < b$. Therefore, there exist such q and r.

(Uniqueness). Suppose that $a = bq_1 + r_1$ and that $a = bq_2 + r_2$, where $0 \leq r_1 < b$ and $0 \leq r_1 < b$. Without loss of generality, we may assume that $r_1 \leq r_2$. This means, $0 \leq r_2 - r_1 \leq r_2 < b$. However, we also have

$$\begin{aligned} 0 \leq r_2 - r_1 &= (a - bq_2) - (a - bq_1) \\ &= b(q_1 - q_2). \end{aligned}$$

That is, $r_2 - r_1$ is nonnegative multiple of b that is less than b. Thus, $r_2 - r_1 = 0$, and hence $r_1 = r_2$. Also, it follow that $bq_1 = bq_2$, where $b > 0$. This implies that $q_1 = q_2$. Therefore, we have shown that $r_1 = r_2$ and that $q_1 = q_2$, and this shows that r and q are unique. Therefore the Theorem. ■

Theorem 1.4 (The Strengthened Division Algorithm) .
Let a and b be integers with $b > 0$. Then there exist unique integers q and r such that

$$a = bq + r \text{ with } 0 \leq r < |b|.$$

Proof. Left as an exercise. ■

1.2 Prime Factors

Def 1.2 (Greatest Common Divisor (GCD)) .
An integer d is a **greatest common divisor** of a and b if the following conditions are hold:

1. $d \in \mathbb{Z}^+$
2. $d \mid a$ and $d \mid b$
3. $(c \mid a) \wedge (c \mid b) \implies c \mid d$.

Theorem 1.5 .
Let a and b be integers, at least one of them is not 0. Then there exists a unique greatest common divisor d of a and b. Moreover, d can be written as a linear combination of a and b, that is,

$$d = am + bn$$

for $m, n \in \mathbb{Z}$, and d is the smallest positive integer that can be written in this form.

Proof. (Existence). Let a and b be integers, at least one of them is not 0. If $b = 0$, then $a \neq 0$, and hence $|a| > 0$. It is obvious that $d = |a|$ is a gcd of a and b in this case, and either $d = a \cdot (1) + b \cdot (0)$ or $d = a \cdot (-1) + b \cdot (0)$.

Suppose now that $b \neq 0$. Consider the set S of all integers that can be written in the form $ax + by$ for some integers x and y, and let S^+ be the set of all positive integers in S. The set S contains $b = a \cdot (0) + b \cdot (1)$ and $-b = a \cdot (0) + b \cdot (-1)$, hence S^+ is not empty. Then by the Well Ordering Principle 1.1, S^+ has a least element d,

$$d = am + bn.$$

We have d positive, and we shall show that d is a gcd of a and b. By the division algorithm 1.3, there are integers q and r such that

$$a = dq + r \text{ with } 0 \leq r < d.$$

Then,

$$\begin{aligned} r &= a - dq \\ &= a - (am + bn)q \\ &= a(1 - mq) + b(-nq). \end{aligned}$$

Thus, r is in S, and $0 \leq r < d$. By choice of d as the least element in S^+, $r = 0$, and hence $d \mid a$. Similarly, it can be shown that $d \mid b$. Now, suppose that $c \mid a$ and $c \mid b$. Then $a = ch$ and $b = ck$, for some integers h and k. Hence,

$$\begin{aligned} d &= am + bn \\ &= chm + ckn \\ &= c(hm + kn), \end{aligned}$$

which shows that $c \mid d$. Hence, by definition 1.2, $d = am + bn$ is the greatest common divisor of a and b. And from the choice of d as least element of S^+ that d is the smallest positive integer that can be written in this form.

(Uniqueness). Left as an exercise. ■

Theorem 1.6 (The Euclidean Algorithm) .
Let a and b be nonnegative integers. Then $gcd(a, b) = r_n$ is the last nonzero remainder after the following successive process:

$$\begin{aligned} a &= bq_0 + r_1, & 0 \leq r_1 < b \\ b &= r_1 q_1 + r_2, & 0 \leq r_2 < r_1 \\ r_1 &= r_2 q_2 + r_3, & 0 \leq r_3 < r_2 \\ &\cdot \\ &\cdot \\ &\cdot \\ r_k &= r_{k+1} q_{k+1} + r_{k+2} & 0 \leq r_{k+2} < r_{k+1}. \end{aligned}$$

Proof. ? - Why we always end up with 0, but not the other nonnegative integers. ■

Def 1.3 (Coprime) .
Two integers a and b are called **coprime** or **relatively prime** if their greatest common divisor is 1.

Theorem 1.7 .
If a and b are relatively prime and $a \mid bc$, then $a \mid c$.

Proof. Suppose that $gcd(a, b) = 1$ and $a \mid bc$. Since $gcd(a, b) = 1$, there exist integers m and n such that

$1 = am + bn$, by Theorem 1.5. And since $a \mid bc$, then by definition there exists an integer q such that $bc = aq$. Now,

$$\begin{aligned} 1 = am + bn &\implies c = amc + bnc \\ &\implies c = amc + aqn \text{ since } bc = aq \\ &\implies c = a(mc + qn) \\ &\implies a \mid c. \end{aligned}$$

Therefore the Theorem. ■

Def 1.4 (Prime Integer) .
An integer p is called **prime integer** if $p > 1$ and the only divisors of p are ± 1 and $\pm p$.

Remark .
Note that the condition $p > 1$ makes p positive and ensures that $p \neq 1$. The exclusion of 1 from the set of primes makes possible the statement of Unique Factorization Theorem.

Lemma 1.2 (Euclid's Lemma) .
If p is a prime and $p \mid ab$, then either $p \mid a$ or $p \mid b$.

Proof. Suppose that p is a prime and $p \mid ab$. If $p \mid a$, then there is nothing to show. Suppose then that $p \nmid a$. This implies that $gcd(p, a) = 1$, since the only positive divisors of p are 1 and p. Then by Theorem 1.7, $p \mid b$. Therefore the Theorem. ■

Corollary 1.7.1 .
If p is a prime and $p \mid (a_1 a_2 \dots a_n)$, then p divides some a_i.

Theorem 1.8 (Unique Factorization Theorem) .
Every positive integer n either is 1 or can be expressed as a product of prime integers, and this factorization is unique, except for the order of the factors.

Proof. (Existence). Let P_n be the statement that either $n = 1$ or n can be expressed as a product of primes. We shall prove that the statement P_n is true for all $n \in Z^+$ by the Second Principle of Finite Induction. P_1 is trivially true. Suppose that P_m is true for all positive integers $m < k$. If k is a prime, then k is a product with one prime factor, and hence P_k is true. Suppose k is not a prime. Then $k = ab$, where neither a or b is 1. Therefore, $1 < a < k$ and $1 < b < k$. By the induction hypothesis, P_a and P_b is true. That is, $a = p_1 p_2 \dots p_r$ and $b = q_1 q_2 \dots q_r$, for primes p_i and q_i. Then,

$$\begin{aligned} k &= ab \\ &= p_1 p_2 \dots p_r \cdot q_1 q_2 \dots q_r, \end{aligned}$$

and k is thereby expressed as a product of primes. thus P_k is true, and therefore P_n is true for all positive integers n. (Uniqueness). Suppose that $n = p_1 p_2 \dots p_t$ and $n = q_1 q_2 \dots q_v$ are factorizations of n as products of prime factors p_i and q_j. Then,

$$p_1 p_2 \dots p_t = q_1 q_2 \dots q_v,$$

So $p_1 \mid (q_1 q_2 \dots q_v)$. By corollary 1.7.1, $p_1 \mid q_j$ for some j, and without loss of generality we may assume $j = 1$. However, p_1 and q_1 are primes, so $p_1 \mid q_1$ implies $q_1 = p_1$. This gives us,

$$p_1 p_2 \dots p_t = p_1 q_2 \dots q_v,$$

and therefore,

$$p_2 \dots p_t = q_2 \dots q_v.$$

By repeating with this process we have finally obtain,

$$p_t = q_t \dots q_v.$$

Since the only positive factors of p_t are 1 and p_t, and since q_j is prime, this means that there must be only one q_j on the right-hand side. Hence $v = t$ and $q_t = p_t$. This completes the proof. ■

Theorem 1.9 (Euclid's Theorem on Primes) .
There are infinitely many primes.

Proof. We shall give the proof by contradiction. Suppose that there are a finite number n of primes, $p_1, p_2, \dots p_n$. Now, consider the integer $m = p_1 p_2 \dots p_n + 1$. It is clear that the remainder in the division of m by any prime p_i is 1, so each p_i is not a factor of m. Thus, there are two possibilities, either m is itself a prime, or it has a prime factor that is different from every one of the p_i. In either case, we have a prime integer that was not in the list. Therefore, there are more than n primes and this completes the proof. ■

1.3 Congruence of Integers

Def 1.5 .
Let n be a positive integer, $n > 1$. For integers x and y, x is said to be **congruent** to y **modulo** n if $x - y$ is a multiple of n, and we write

$$x \equiv y \pmod{n}.$$

Remark .
Note that $x \equiv y \pmod{n} \iff x - y = nq$ for some $q \in \mathbb{Z}$.

Theorem 1.10 .
The relation of congruence modulo n is an equivalence relation on $\mathbb{Z}$.

Reflexive.

$x \equiv x \pmod{n}$ since $x - x = (n)(0)$.

Symmetric.

$$\begin{aligned} x \equiv y \pmod{n} &\implies x - y = nq \quad \exists q \in \mathbb{Z} \\ &\implies y - x = n(-q), \ -q \in \mathbb{Z} \\ &\implies y \equiv x \pmod{n}. \end{aligned}$$

Transitive.

$$\begin{aligned} x \equiv y \pmod{n} \text{ and } \quad & y \equiv z \pmod{n} \\ &\implies x - y = nq,\ y - z = nk \\ &\exists q, k \in \mathbb{Z} \\ &\implies x - z = x - y + y - z \\ &= n(q + k), ,\ q + k \in \mathbb{Z} \\ &x \equiv z \pmod{n}. \end{aligned}$$

Remark .
Remember that any equivalence relation, the equivalence classes for congruence modulo n form a *partition* of $\mathbb{Z}$; that is they separate $\mathbb{Z}$ into mutually disjoint subsets. These subsets are called **congruence classes**. For example, when $n = 4$, the congruence classes appears as:

$$\begin{aligned} [0] &= \{\ldots, \quad -4, \quad 0, \quad 4, \quad \ldots\} \\ [1] &= \{\ldots, \quad -5, \quad 1, \quad 5, \quad \ldots\} \\ [2] &= \{\ldots, \quad -6, \quad 2, \quad 6, \quad \ldots\} \\ [3] &= \{\ldots, \quad -7, \quad 3, \quad 7, \quad \ldots\} \end{aligned}$$

Theorem 1.11 .
If $a \equiv b \pmod{n}$ and x is any integer, then $a + x \equiv b + x \pmod{n}$ and $ax \equiv bx \pmod{n}$.

Proof. Left as an exercise. ■

Remark .
We can use cancellation on addition but not on multiplication. By this I mean,

$$a + x \equiv a + y \pmod{n} \implies x \equiv y \pmod{n},$$

but

$$ax \equiv ay \pmod{n} \not\Longrightarrow x \equiv y \pmod{n}.$$

The second one can be easily shown by giving a counter-example.

Theorem 1.12 (Cancellation Law) .
If $ax \equiv ay \pmod{n}$ and $gcd(a, n) = 1$,
then $x \equiv y \pmod{n}$.

Proof. Suppose $ax \equiv ay \pmod{n}$ and $gcd(a, n) = 1$. Then,

$$\begin{aligned} ax \equiv ay \pmod{n} &\implies n \mid (ax - ay) \\ &\implies n \mid a(x - y) \\ &\implies n \mid (x - y) \text{ by Theorem 1.7} \\ &\implies x \equiv y \pmod{n}. \end{aligned}$$

This completes the proof. ■

Theorem 1.13 (Substitution Properties) .
Suppose $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$.
Then, $a + c \equiv b + d \pmod{n}$ and $ac \equiv bd \pmod{n}$.

Proof. Left as an exercise. ■

Ex. 1.3.1. Find the integer x for $58^{23} \equiv x \pmod{9}$.

Solution. Since $58 \equiv 4 \pmod{9}$, then by Theorem 1.13, we have $58^{23} \equiv 4^{23} \pmod{9}$. Also since $4^{23} = 4^2 \cdot (4^3)^7$, then

$$\begin{aligned} 58^{23} &\equiv 4^{23} \pmod{9} \\ &\equiv 4^2 \cdot (4^3)^7 \pmod{9} \\ &\equiv (16)(64)^7 \pmod{9} \\ &\equiv (7)(1)^7 \pmod{9} \\ &\equiv 7 \pmod{9}. \end{aligned}$$

This solves the problem. □

Theorem 1.14 (Linear Congruences) .
If a and n are co-prime, the congruence $ax \equiv b \pmod{n}$ has a solution x in the integers, and any two solutions in $\mathbb{Z}$ are congruent to modulo n.

Proof. Since a and n are co-prime, there exist integers s and t such that

$$\begin{aligned} 1 &= as + nt \\ &\implies b = asb + ntb \\ &\implies a(sb) - b = n(-tb) \\ &\implies n \mid [a(sb) - b] \\ &\implies a(sb) \equiv b \pmod{n}. \end{aligned}$$

Thus, $x = sb$ is a solution to $ax \equiv b \pmod{n}$.

(Uniqueness). To complete the proof, suppose that both x and y are integers that are solutions to $ax \equiv b \pmod{n}$. Then we have,

$$ax \equiv b \pmod{n} \quad \text{and} \quad ay \equiv b \pmod{n}.$$

Then by symmetric and transitive properties of congruence modulo n, we conclude that these relations imply

$$ax \equiv ay \pmod{n}.$$

Since $gcd(a, n) = 1$, then by Cancellation Law 1.12 we get $x \equiv y \pmod{n}$. Therefore, any two solutions in $\mathbb{Z}$ are congruent modulo n. ■

Theorem 1.15 .
Let m and n be co-prime and a and b be integers. There exists an integer x that satisfies the system of congruences

$$\begin{aligned} x &\equiv a \pmod{m} \\ x &\equiv b \pmod{n}. \end{aligned}$$

Furthermore, any two solutions x and y are congruent modulo mn.

Proof. (Existence). Suppose $gcd(m, n) = 1$. Let x be a solution to the first congruence in the system

$$\begin{aligned} x &\equiv a \pmod{m} \\ x &\equiv b \pmod{n}. \end{aligned}$$

Thus, $x = a + mk$ for some integer k, and this must be such that $a + mk \equiv b \pmod{n}$ or, $mk \equiv b - a \pmod{n}$. Since $gcd(m, n) = 1$, Theorem 1.14 guarantess the existence of such an integer k, and $x = a + mk$ satisfies the system.

(Uniqueness). Left as an exercise. ■

Theorem 1.16 (Chinese Remainder Theorem) .
Let $n_1, n_2, \ldots n_m$ be pairwise co-prime. There exists an integer x that satisfies the system of congruences

$$\begin{aligned} x &\equiv a_1 \pmod{n_1} \\ x &\equiv a_2 \pmod{n_2} \\ &. \\ &. \\ &. \\ x &\equiv a_m \pmod{n_m}. \end{aligned}$$

Furthermore, any two solutions x and y are congruent modulo $n_1 n_2 \ldots n_m$.

Proof. Left as an exercise. ■

Theorem 1.17 (Fermat's Little Theorem) .
If p is a prime number, then for any integer a we have

$$a^p \equiv a \pmod{p}.$$

Proof. **RETURN!** ■

1.4 Congruence Classes

We have observed that, in connection with the relation of congruence modulo n, there are n distinct congruence classes. Let $\mathbb{Z}_n$ denote this set of classes:

$$\mathbb{Z}_n = \{[0], [1], \ldots [n-1]\}.$$

Theorem 1.18 .
Consider the rule given by

$$[a] + [b] = [a + b].$$

Then,

a. This rules defines an addition that is a binary operation on $\mathbb{Z}_n$.

b. Addition is associative in $\mathbb{Z}_n$.

c. Addition is commutative in $\mathbb{Z}_n$.

d. $\mathbb{Z}_n$ has the additive identity $[0]$.

e. Each $[a]$ in $\mathbb{Z}_n$ has $[-a]$ as its additive inverse in $\mathbb{Z}_n$.

Proof.

a. It is clear that the rule $[a]+[b]=[a+b]$ yields and element of $\mathbb{Z}_n$. In other words, closure is obvious, but we need to show that the operation is well-defined. To do this, suppose that $[a]=[x]$ and $[b]=[y]$. Then $[a]=[x] \implies a \equiv x \pmod{n}$ and $[b]=[y] \implies b \equiv y \pmod{n}$. Then by Theorem 1.13, $a+b \equiv x+y \pmod{n}$, and therefore $[a+b]=[x+y]$. Therefore, the rule is well-defined.

b.

$$\begin{aligned}[a]+([b]+[c]) &= [a]+[b+c] \\ &= [a+(b+c)] \\ &= [(a+b)+c] \\ &= [a+b]+[c] \\ &= ([a]+[b])+[c].\end{aligned}$$

c. Left as an exercise.

d. Since addition is commutative in $\mathbb{Z}_n$ and $[a]+[0]=[a+0]=[a]$.

e. Left as an exercise.

■

Remark .
Note that the key step in part b. of the proof is the fact that addition is associative in $\mathbb{Z}$.

Theorem 1.19 .
Consider the rule for multiplication in $\mathbb{Z}_n$ given by

$$[a][b]=[ab].$$

Then,

a. Multiplication as defined by this rule is a binary operation on $\mathbb{Z}_n$.

b. Multiplication is associative in $\mathbb{Z}_n$.

c. Multiplication is commutative in $\mathbb{Z}_n$.

d. $\mathbb{Z}_n$ has the multiplicative identity $[1]$.

Proof. Left as an exercise. ■

Remark .
A multiplication table for $\mathbb{Z}_4$ is shown below. The third row of the table shows that $[2]$ is a nonzero element of $\mathbb{Z}_4$ that has no multiplicative inverse. Another interesting point is that the equality $[2][2]=[0]$ shows that the product of nonzero factors may be zero.

$\times$	$[0]$	$[1]$	$[2]$	$[3]$
$[0]$	$[0]$	$[0]$	$[0]$	$[0]$
$[1]$	$[0]$	$[1]$	$[2]$	$[3]$
$[2]$	$[0]$	$[2]$	$[0]$	$[2]$
$[3]$	$[0]$	$[3]$	$[2]$	$[1]$

Theorem 1.20 .
An element $[a]$ of $\mathbb{Z}_n$ has a multiplicative inverse in $\mathbb{Z}_n$ if and only if a and n are co-prime.

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1.5 Exercises

Ex. 1.5.1.

a. Prove that $(\forall n \in \mathbb{Z}^+)10^n \equiv 1 \pmod 9$.

b. Prove that a positive integer is divisible by 9 if and only if the sum of its digits is divisible by 9.

Ex. 1.5.2.

a. Prove that $(\forall n \in \mathbb{Z}^+)10^n \equiv (-1)^n \pmod{11}$.

b. Prove that a positive integer z is divisible by 11 if and only if 11 divides $a_0 - a_1 + a_2 - \cdots + (-1)^n a_n$, where $z = a_n 10^n + a_{n-1}10^{n-1} + \cdots + a_1 10 + a_0$.

BIBLIOGRAPHY